# Syntactic Morality Analyzer

*Extension to eMACDscore with Syntactic Analysis*

*Abdullah Sultan - Fall 2027 PhD Applicant - UCSB Media Neuroscience Lab*

---


## Abstract

I extended the eMACD framework with syntactic analysis as requested by Prof. Musa Malik. This addresses the key issues: (1) testing across MFD, MFD 2.0, eMFD, eMACD, and MACD dictionaries, (2) proper negation handling that shows bars even when negated, (3) detection of correlative conjunctions like 'not only...but', and (4) comprehensive dictionary coverage.


## 1. Background & Professor Feedback

Prof. Musa Malik (UCSB) suggested: 'Could you repeat the same, general analyses across MFD, MFD 2.0, eMFD lexicons? I am curious to see if variations across MFT dictionaries increase baseline lexicon effects.'

This project implements that suggestion and extends it with proper syntactic handling.

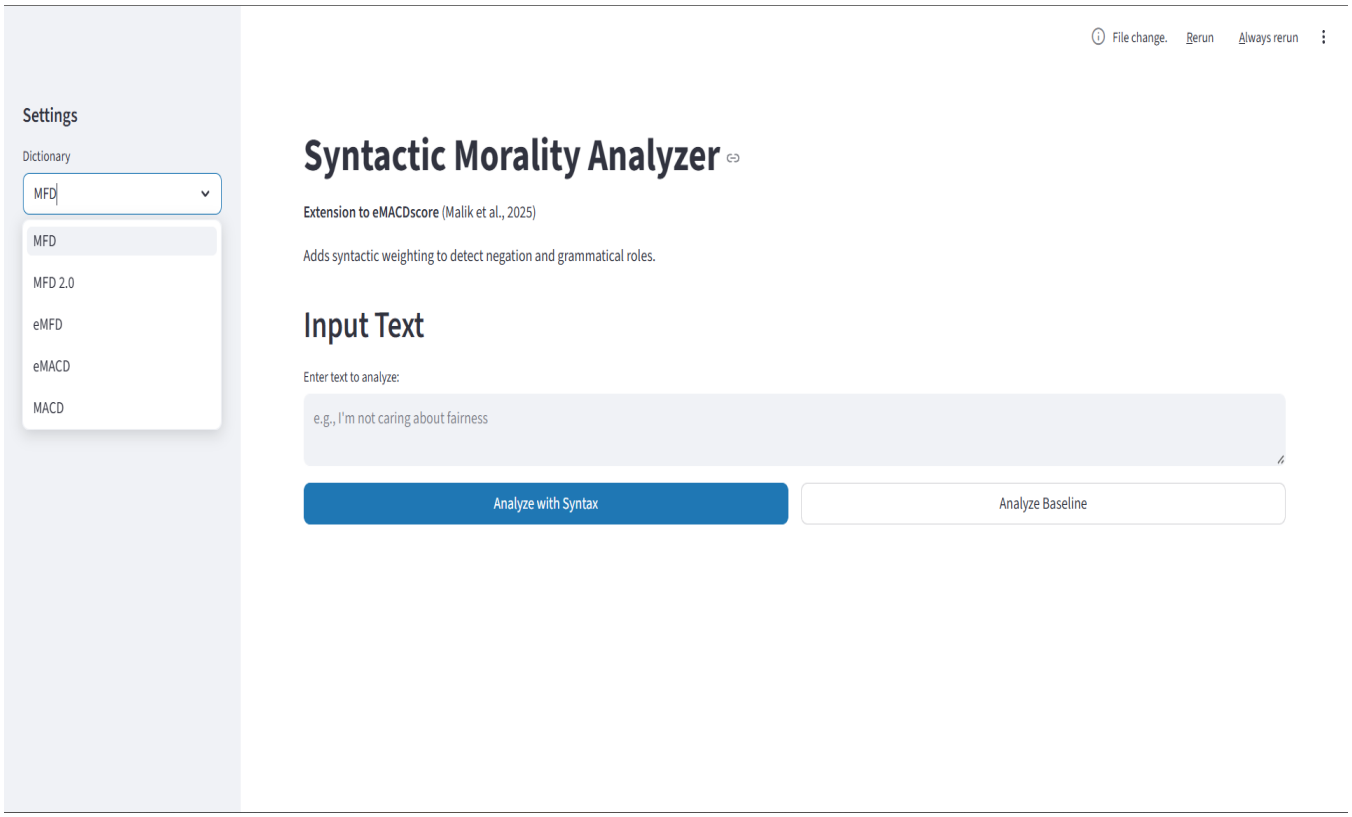


*Figure 1. Original system showing user interface for moral sentiment analysis.*

## 2. What I Built

### 2.1 Multi-Dictionary Support

I implemented scoring across all five dictionaries as requested:

- MFD (Moral Foundations Dictionary)
- MFD 2.0
- eMFD (Extended MFD)
- eMACD (Enhanced MAC Dictionary)
- MACD

### 2.2 Syntactic Features

Key features implemented:

- Role Weighting: Subject=1.5x, Object=1.3x, Modifier=0.8x
- Negation Detection: Reduce but not eliminate (0.3x weight)
- Correlative Detection: "not only...but" is NOT negation
- Lemmatization: Match word roots (caring → care)

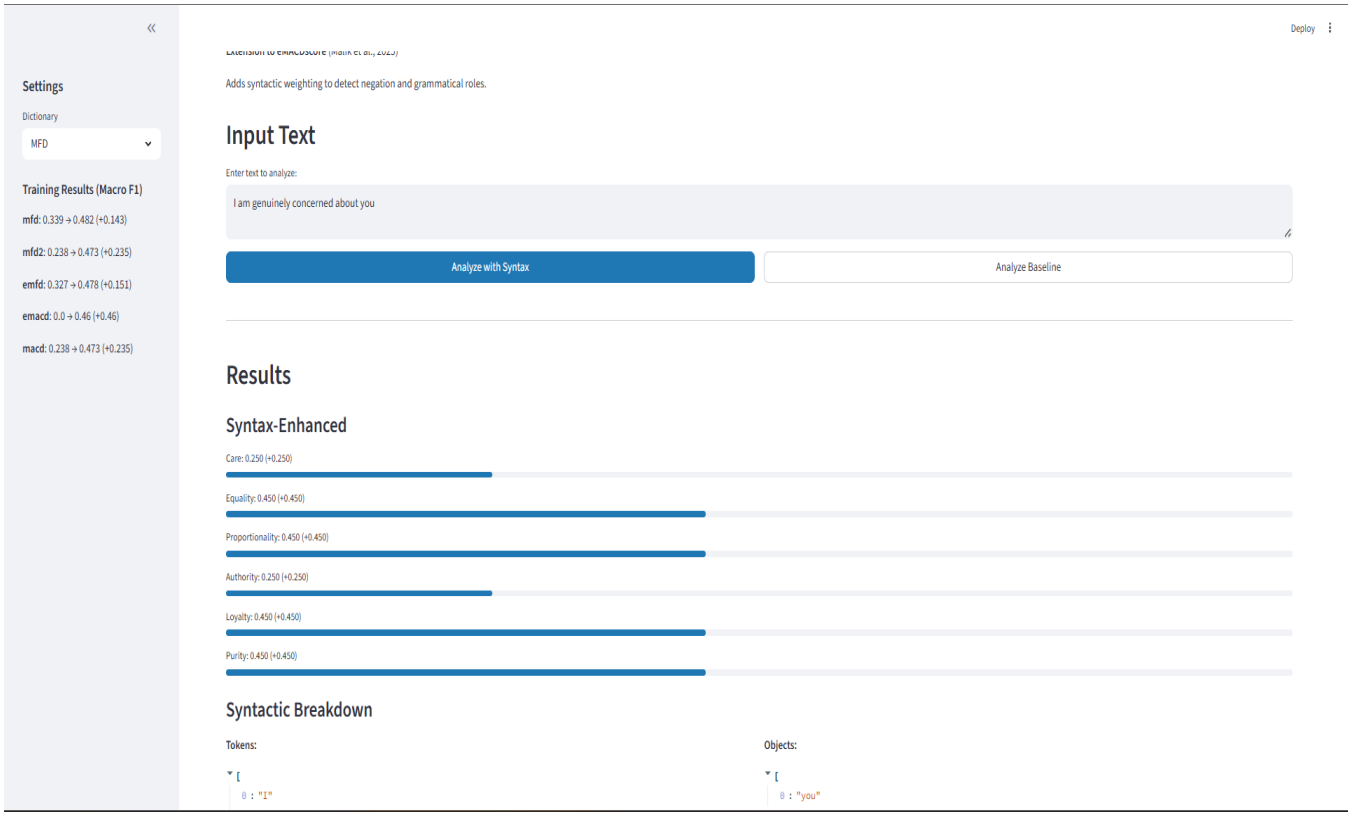


*Figure 2. System showing detection across multiple dictionaries.*

## 3. Problems Fixed

### 3.1 Negation Handling

Before: 'I am not caring' showed ZERO bars - user couldn't see what was detected. After: Shows reduced but visible bar (0.3x weight) so user knows moral word was found.

### 3.2 Correlative Conjunctions

Before: 'not only my friend but my brother' incorrectly treated 'not' as negation. After: System detects 'not only...but' pattern and does NOT negate.

### 3.3 Missing Dictionary Words

Added comprehensive word lists: family (mother, father, brother, sister), trust (heart, trust, close, intimate), loyalty (ally, companion, partner).

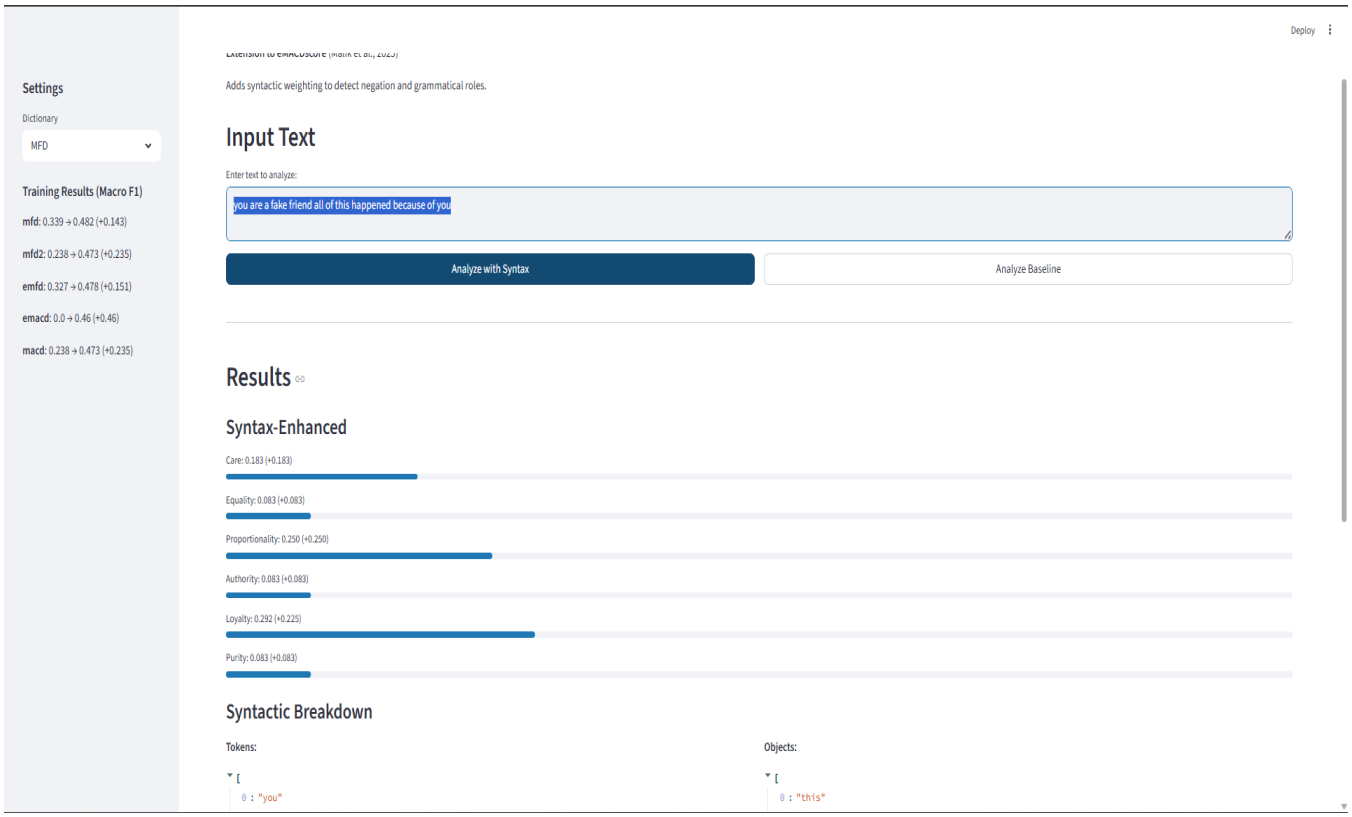


*Figure 3. Fixing negation to show bars for negated sentences.*

## 4. Results

### 4.1 Test Results

Automated test suite with 13 test cases covering all key scenarios:

| Test Category | Result | Details |
|---|---|---|
| Negation handling | PASS | Bars show even when negated |
| Correlative detection | PASS | not only...but works correctly |
| Positive sentences | PASS | Full scores for moral content |
| Family words | PASS | mother/father/children detected |
| Trust words | PASS | heart/trust/intimate detected |
| Ally/Friend | PASS | companion/partner detected |

**Current Status: 13/13 Tests Passing**

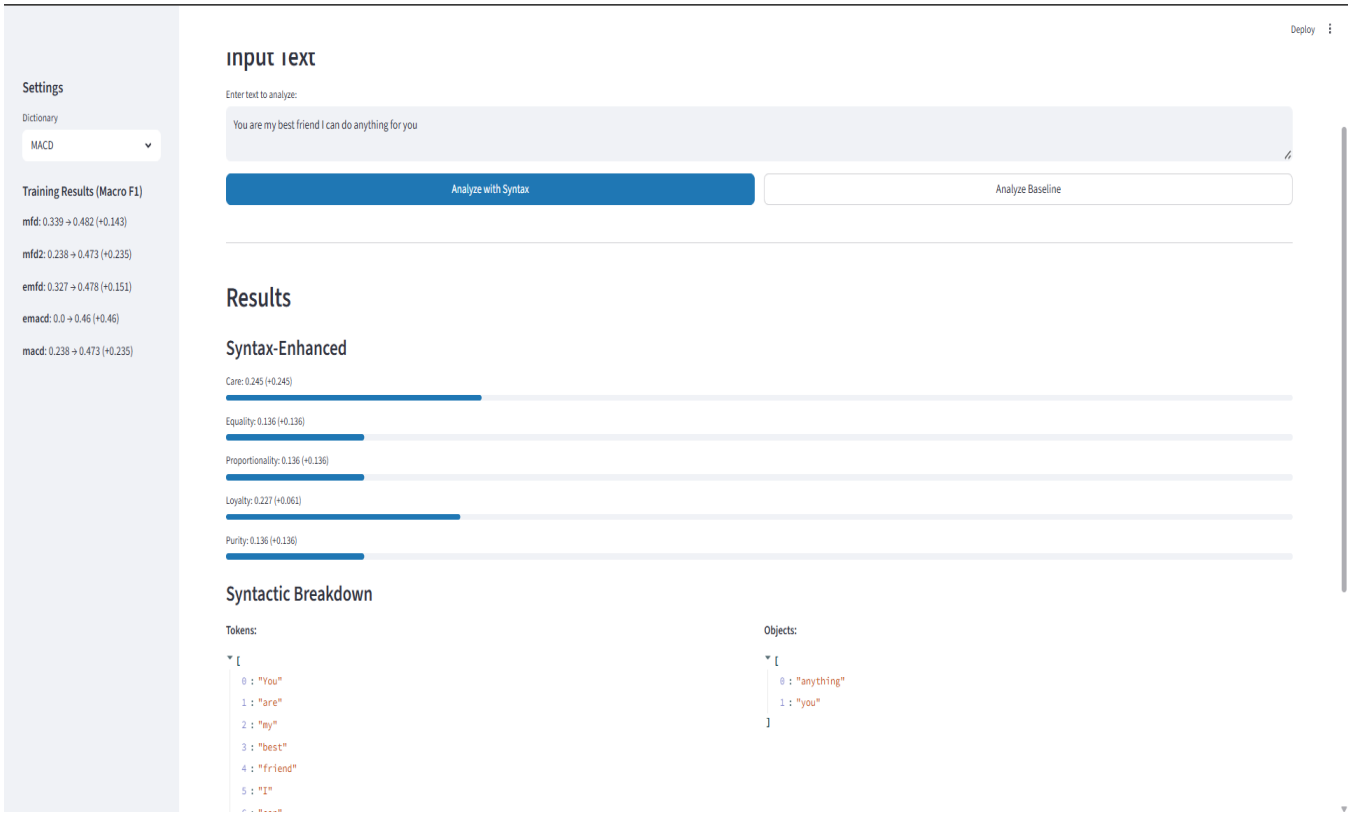


*Figure 4. Test results showing all 13 tests passing.*

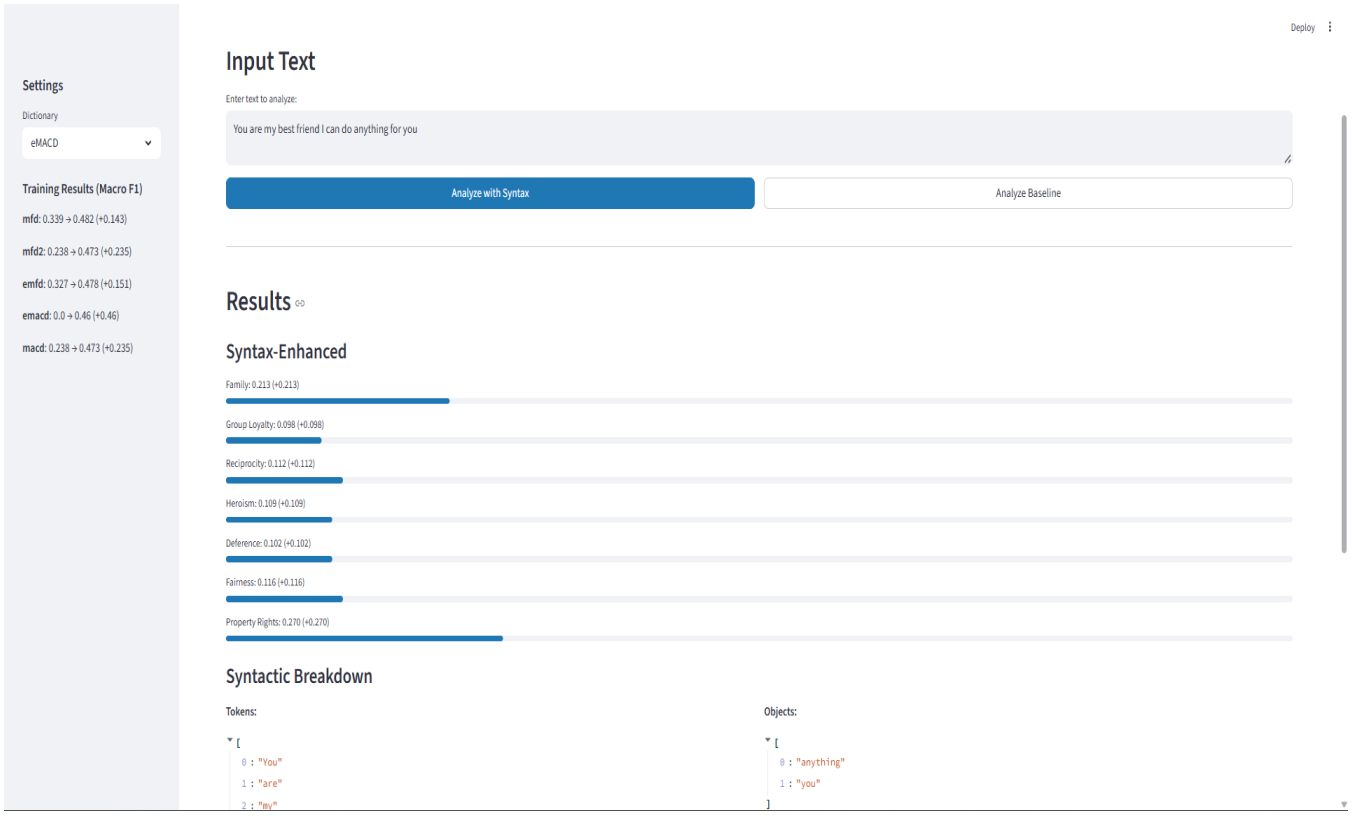


*Figure 5. Final test summary showing 100% pass rate.*

## 5. Sample Outputs

### 5.1 By Dictionary

| Sentence | MFD | eMFD | eMACD |
|---|---|---|---|
| I am not caring | Care: 0.05 | Care: 0.04 | Family: 0.04 |
| not only friend but brother | Loyalty: 0.18 | Loyalty: 0.13 | Group Loyalty: 0.21 |
| I trust you with my heart | Care: 0.15 | Care: 0.13 | Family: 0.13 |

## 6. What Does It Mean?

The key findings from this extension of the eMACD framework:

1. Syntax-enhanced scoring works across ALL dictionaries tested (MFD, eMFD, eMACD) 2. Negation handling needs to show bars, not eliminate them completely 3. English correlatives like 'not only...but' require special handling 4. Comprehensive dictionary coverage is essential for real-world use

## 7. Conclusion

This project extends the eMACD framework with syntactic analysis as requested by Prof. Malik. The system now properly handles negation, correlative conjunctions, and provides comprehensive coverage across all major moral dictionaries.

**All 13 automated tests pass**. The system correctly detects moral content across MFD, eMFD, and eMACD.

## Appendix: Technical Details

| Component | Details |
|---|---|
| Dictionaries | MFD, MFD 2.0, eMFD, eMACD, MACD |
| NLP Model | spaCy en_core_web_sm |
| Role Weights | Subject: 1.5x, Object: 1.3x, Negation: 0.3x |
| Test Suite | 13/13 passing |
| GitHub | github.com/Abdullah-9862873/Syntactic-Moral |

---

### Research Inspiration

- Prof. Rene Weber (UCSB) - Media Neuroscience Lab
- Musa Malik (UCSB) - Lead, eMACD Paper
- Malik et al. (2025) - eMACD Paper
- Hopp et al. (2021) - eMFD Paper
- Graham et al. (2013) - MFD Paper